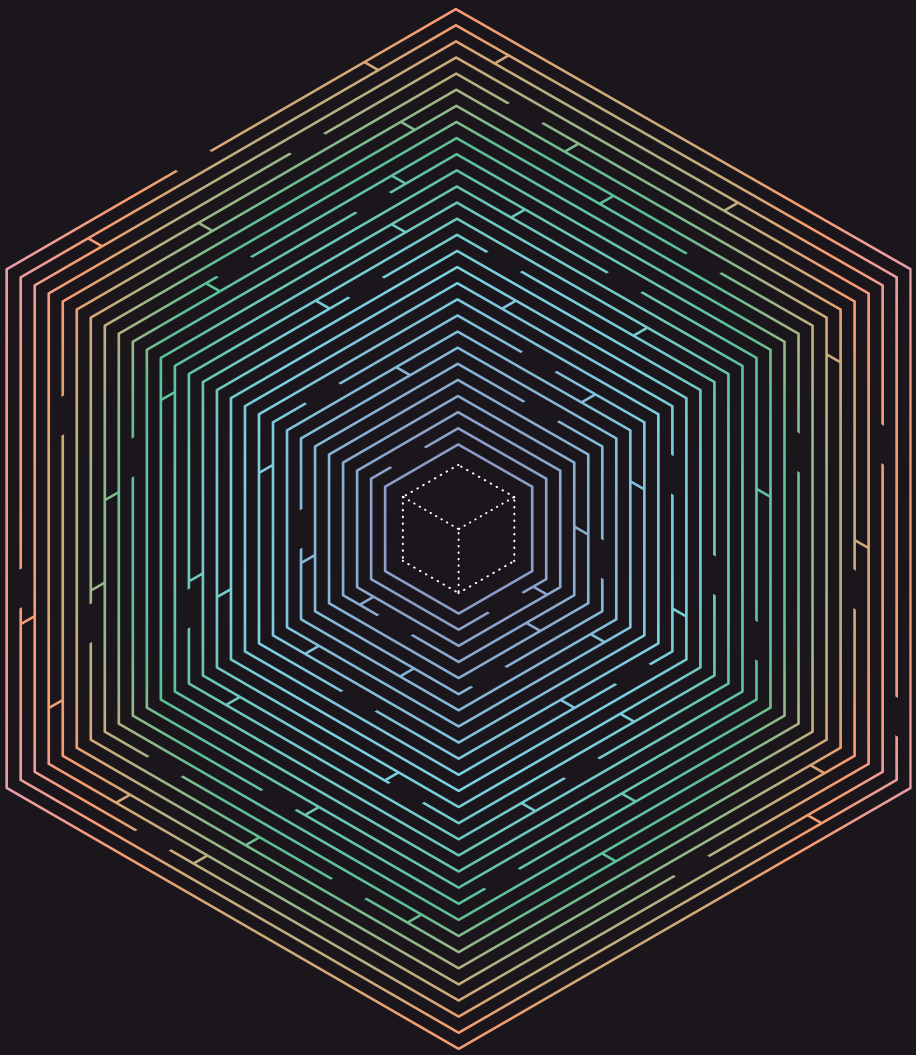
# Age of Invisible Machines

A Practical Guide to Creating a Hyperautomated Ecosystem of Intelligent Digital Workers

Robb Wilson
with Josh Tyson



WILEY

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# PREFACE

By Josh Tyson

*I came to Age of Invisible Machines with a solid understanding of the technologies involved in hyperautomation but only a vague notion of how they fit together. Most of this knowledge was seeded in 2012, when I began working as managing editor of UX Magazine, the world's longest-running publication for the experience design community. And Robb Wilson, as the owner of UXM, had a mythical aura about him.*

*Given that Robb is a pioneering force in the burgeoning field of experience design, colleagues talked about his ideas and achievements with a mix of awe and affection. Robb had written an early book on effective interface design and had been tapped by Apple to create one the first iPad apps. He'd already been refining his own conversational AI machine named Cybil for more than 10 years. Apparently, he'd even been nominated for an Academy Award from his work in film. Bursting with big ideas that came rolling down the pike at a steady clip, he was running multiple international startups and divided his time between Denver and Kyiv. I rarely saw Robb, but he was always there.*

*A couple of years ago, I seized an opportunity to spelunk the elusive Robb mind, helping to refine a whitepaper about hyperautomation that quickly mushroomed into this book. I was astounded, not only by the complexity and scope of hyperautomation as Robb describes it, but also by the platform he's created for building ecosystems where hyperautomation can thrive. Imagine the power of an open system that connects all of the data points (and data stores) within an organization. Anyone can use the platform to sequence a host of disruptive technologies without having to write code, which means anyone can leverage data and create better-than-human automations at every level of society via an interface—conversation—that requires no training.*

*Named by the tech research and consulting firm Gartner, "hyperautomation" is an inevitable market state and also a convoluted, hairy affair. The deeper I've delved into it, the more I've noticed similar systems in nature. On a hike, for instance, I'll notice how the elegant balance of the individual elements in a healthy forest—the spongy moss, the immovable stones, towering trees—creates a wholly immersive experience made possible by an invisible network sprawling underfoot in every direction. Called "mycelium," this fungal network connects 92% of plant families. The symbiotic relationship between mycelium and plants is called "mycorrhiza"; plants provide the fungi with sugars while the fungi transfers nutrients and water from the soil. The mycorrhiza runs through a network that can spread for miles. All of this goes largely unnoticed—but without it, there'd be no splendor.*

*Ecosystems for achieving and maintaining hyperautomation are similar. A complex network of*

*The pandemic has sealed the trend of people's primary interactions with organizations—as both customers and employees—being digital in nature. According to Gartner, businesses can gain a sizable competitive advantage in the era of remote work and distributed customers by combining multi-experience, customer experience, employee experience, and user experience into something they call total experience (TX). Using conversational interfaces can maximize impact across a full spectrum of experiences and greatly assist efforts to achieve hyperautomation. [NOTE "Why 'Total Experience (TX)' is Gartner's top technology trend for 2022," Clover Infotech, December 22, 2021, https://www.cloverinfotech.com/blog/why-total-experience-tx-is-gartners-top-technology-trend-for-2022/]*

*In this regard, conversational AI represents a massive step forward. If users can simply converse with technology, the pain points designers continually have to wrangle with disappear. Of course, the complexity underfoot grows considerably. We'll explore both facets of this inverse relationship: what's possible through achieving hyperautomation and the massive amount of orchestration required to bring those possibilities to life.Even though a few companies are already achieving, maintaining and accelerating hyperautomation, much of what Robb describes in this book is still on the verge of taking flight. Similarly, as we worked on the book, the finish line would often recede, the topic ever expanding. There's so much to cover, and the learning process can be very different from the doing process. I've come to think of this book as more of a living document. It's the completed product you're reading now, but it's also connected to a growing library of information, resources, and tools available online at invisiblemachines.onereach.ai or via the QR codes you'll find throughout the text.*

*No matter what your level of experience is with this kind of technology, make no mistake that this is a clarion call announcing an inevitable and rapidly approaching evolution that will upend our relationship with machines. Our goal is to give you the perspective, practical advice, and strategy necessary to remain on firm ground as everything around us changes in ways we're still imagining. These changes are already underway, and the leaders of tomorrow are hard at work today, bringing their organizations closer to hyperautomation.*

# INTRODUCTION

***Conversational AI, My White Whale By Robb Wilson***

*Call me Ishmael.*

*Actually, please don't.*

*Like Captain Ahab in Moby Dick, I've spent many a waking hour in heated pursuit of a powerful and elusive white whale: conversational AI. For lingering days, months, and years I've chased this steely beast on the horizon. I was frequently knocked off course by the complexity and newness of the various associated technologies, but I kept up the chase on both sides of land and over all sides of earth (as Melville would say).*

*For me, this obsession began as an early practitioner in the field of experience design. I noticed that the absolute worst experiences people were routinely having with machines were conversational in nature: purgatorial voice automated call centers; feeble chatbots trying to solve problems online and wreaking havoc on user trust with their inefficiency. Lifting users and organizations out of the seemingly infinite shitbot doldrums seemed like the true white whale of experience design.*

*It was easy to understand why much of the focus in experience design has been put on creating graphical UIs that draw users in and give them rewarding experiences—wherein complex interactions with machines were easier to manage. Our interactions with computers have been screen based for ages; and as those screens have reached higher definition, been reduced in size, and become touch activated, new opportunities to empower and delight have emerged. I've seen some impressive graphical UIs over the years, but there's a ceiling on how much complexity you can condense into a purely visual interface.*

*On the other hand, in exploring conversational interfaces I realized that nothing could make interactions with machines easier than using our most natural forms of communication. Speech and text are methods for sharing information that nearly everyone on the planet can leverage without special training. There's no need for users to scour a top navigation or figure out what certain icons mean if they can simply converse with the machine functioning underneath the cumbersome GUI. It was clear to me that, if conversational AI could fulfil its promise, it would effectively mask the machinations, making the machine invisible.*

*<INSERT LINE BREAK>*

*I've now been talking about talking to machines for decades. I recently found myself having a familiar conversation over coffee with the CEO of a publicly traded tech company. Once again, I was hitting on the ways conversational AI and, more broadly, hyperautomation will destroy the status quo.*

*"Putting technology to work will be as simple as asking for its help."*

*"Using conversation to engage machines' vast problem-solving powers can literally make anyone a technologist."*

*"Humans will be able to leverage advanced technology without a formal understanding of how to create and deploy software."*

*"Eventually, we'll even put technology to work by having software build new software."*

*The CEO smiled and said it all sounded amazing, but I could tell the immediacy wasn't hitting home. This person helms a company that, at the time of our conversation, was managing projects to build machines for Slack and Teams so as to make their own apps available through those conversational interfaces. I was left waiting to see if he'd realize that his organization is literally undermining their own platform's graphical UI in an effort to accommodate a conversational one.*

*There have been attempts to scale GUIs, like Sharepoint and Salesforce, but they ultimately revealed the hard truth that a UI with a hundred tabs designed by as many people is even worse to navigate than it sounds. No doubt, one of the biggest reasons Microsoft is moving from Sharepoint to Teams is the scalability of a conversational interface that's connected to everything. Customers and employees alike can interact with a company through one portal that ties together and obscures the sausage factory behind the scenes.*

*Really, the missing link in the creation and widespread adoption of strategies for achieving hyperautomation is a scalable interface. Graphical UIs will never scale to meet the requirements of hyperautomation. This goes well beyond the issue of being too cluttered and overwhelming; it's often the very processes graphical UIs render that need to be automated the most. Conversational AI is the missing link, the infinitely scalable interface. Not only does it handily obscure the mess of systems (and graphical UIs) running behind the scenes, it also binds your ecosystem, creating a feedback loop that can evolve automations for all your users—customers and employees.*

*Companies like Lemonade and Ant Financial are already leveraging conversational AI and hyperautomation to fundamentally alter their respective industries. For this CEO I was conversing with, a frightening realization is waiting around the corner: once wider adoption is resolved, conversational interfaces will undoubtedly scale in ways that will completely obscure graphical UIs.*

*In the course of building a platform for leveraging conversation AI I've come to realize that it's really a combination of three key technologies: conversational user interfaces, composable architecture, and no-code rapid application programming. In essence this trinity is truly the white whale of experience design. It furthers the frictionless conversational interface to a point where software creation becomes democratized. Instead of focusing on the experience of using software, code-free allows us to elevate our efforts to designing the conversational experience of creating software using composable architecture. Not only do these component technologies elevate UX design to new heights, they can do the same for humanity, ensuring everyone has access to technology and can easily put it to use: technology that doesn't leave anyone behind.*

*When the automation of your organization's external and internal operations are leveraged through conversational AI, anyone can create and iterate software solutions on their own. Conversational AI, orchestrated alongside other disruptive technologies on an open platform, can create an ecosystem that allows your team members to work collaboratively on automating tasks and processes in ways that people alone could never achieve. Conversational AI gives customers infinitely more rewarding interactions with your organization while also giving your team members an entirely new paradigm within which to work.*

*The requirements for this level of automation—hyperautomation—create a scenario where the various elements of experience design become part of a rapidly moving feedback loop. For lack of a more creative term, this represents a kind of hyperUX. The energy previously poured into designing around the limitations of graphic UIs now gets poured into designing for an infinitely scalable conversational interface. The architecture of these experiences requires extensive journey mapping, and those maps become living documents that evolve alongside the experiences. Extensive research and analytics can happen in real time as the people inside your organization watch users moving though experiences as they unfold. And, unlike what happens in our current reality when machines inevitability get stuck, the overall hyperUX experience can continually be rewarding because it's designed to have humans jump in and assist whenever needed. This creates an environment that's more agile than agile, where code-free interactions with machines allow for iterations and improvements being made constantly.*

*This might sound far-fetched to anyone who's been held hostage by the sluggish development cycles of their key vendors, but in an ecosystem built for hyperautomating you truly can pull in any piece of technology and make significant changes to any aspect of a user's experience at a moment's notice. I've seen many high-minded projects get dragged through protracted development processes only to stumble down into "release and regret" existence. At its best, building software is a creative process. By applying the strategies and techniques described in this book, you can reach a place where the creative people within your organization can manage the process of software creation from end to end.*

*If you, like me, have also chased after conversational AI, by now you've likely recognized that doing it the wrong way feels eerily like Ahab's fate: his ship split in two by a forceful beast he never properly reckoned with, he finds himself tied to that force and dragged out to sea. I'm still astonished by the fact that I'm now able to swim alongside my white whale. Taken together in a swirling mass, conversational AI and hyperautomation become something beautiful and monstrous and beguiling. This endeavor doesn't amount to pursuing a trophy you win with a hundred harpoons. Those who try to conquer these technologies will find their schemes smashed to bits. The trick is to be nimble and swift, riding the waves alongside them as everything churns madly around you. The prize of 'catching' conversational AI matters so much because of what it symbolizes: you've learned to harness conversational UI, composable architecture, and no-code creation by creating a strategic environment where everyone can use technology effectively and no one is left behind (or out to sea). In the pages to follow, we'll take a deep dive into the complexity of this new realm. By sharing with you everything I've learned, I want to help your organization swim with the speed, strength, and flexibility necessary to propel your organization forward in choppy waters. This will be an arduous and complex journey fraught with difficult decisions, but it will also give you the chance to reflect on the outdated processes and systems that have been holding you back. But whether this sounds appealing to you or horrifying, the bottom line is it's time to take the plunge. In this space I'll share the perspective of my journey over the course of 20+ years. If you're going to do this, it's crucial to have a point of view, so I'll lend you mine.*

PART ONE

# IMAGINING AN ECOSYSTEM OF INTELLIGENT WORKERS

As I've already hinted in the introduction, creating an ecosystem that supports achieving a state of hyperautomation is a monumental task, and you'll want to approach it with solid foundational knowledge. In Part I, we'll take a look at the current landscape (as of this writing), dispel some of the common myths associated with conversational AI, and start to paint a picture of what successfully achieving hyperautomation looks like. With technology this all-encompassing and powerful, it's also crucial to bear in mind the myriad ethical concerns that will arise. These orchestrated technologies will upend our world in rather sudden ways, transforming established chunks of our daily routines. This work must be done with more than just good intentions; we must pay close attention to the shifting outcomes of those good intentions as things begin to accelerate. And accelerate they will. We are more or less at the starting point; from here we can set a trajectory that benefits everyone, leaving no one behind.

Massive opportunities are waiting for the organizations that take these efforts seriously, and that are willing to make the sacrifices of letting go of outdated systems and structures. There is a world within reach where everyone has access to technology and no one is left behind—where soul-sucking jobs are a thing of the past, and companies are increasingly self-driving. In this world, people are free to work together on the most interesting, creative problems, and not only within the confines of large companies. The strategic application of conversational AI to achieve a state of hyperautomation is just a means to that end—to that place where technology benefits everyone equally.

This is the book about how to get to that world. It's about how to get your team and company to organize and operate in ways that are highly conducive to achieving and maintaining a state of hyperautomation, which is what allows your organization to become increasingly self-driving. Being in a state of hyperautomation is not unlike someone in a state of ketosis—having starved a body of carbohydrates to burn for energy so it starts burning fat for fuel instead. Hyperautomation means reorganizing your organization's insides (and likely starving it of outdated tools and processes) so that it can exist in a far more potent and efficient state. To whet this new appetite, let's first explore some scenarios of where hyperautomation can take your organization (and the world).

CHAPTER ONE

# HYPERDISRUPTION IS ALREADY HERE

# HYPERDISRUPTION IS ALREADY HERE

I don't want to scare you, but this book comes with an ominous warning: the status quo is a death sentence. Most modern organizations are run using systems and strategies that will seem almost comically outdated a few short years from now. That's because the strategic orchestration of technologies described in this book—conversational AI, hyperautomation, and intelligent automation—will do a lot more than disrupt the ways we're accustomed to dealing with technology; that strategic orchestration will obliterate existing models.

We're living in an era when technology moves in exponential leaps and bounds, every day growing more powerful, more pervasive, and more sophisticated. It's no coincidence that ninety percent of all the data in existence was created over the past two years.

The technologies surrounding conversational AI are heading toward a point of convergence that will fundamentally alter our relationship with machines. Already, the experiences customers and employees have with businesses are being reshaped by the hallmarks of this convergence—putting those massive stores of data into action in ways that have upended entire industries. This might sound hyperbolic—especially given the substandard chatbot experiences endemic to much of the automation happening in the world at present—but it's not.

The reason that conversational AI in particular feels like a lot of talk is simply because it hasn't been more widely adopted—yet. The technology surrounding conversational AI has become highly sophisticated, but that hasn't changed the fact that people don't like pogo-sticking between ten different machines. For example, let's say you log on to your home security system website to cancel service. Asking their chatbot a question drops you down a funnel of FAQ menus where you learn, five minutes in, that cancelations can't be handled online. When you next call the accounts department, you're confronted by a series of voice automations that feels like another funnel drop, so you start stamping "0" hoping for a shortcut to a live agent. Crap experiences like this can feel less productive than just waiting on hold and hoping you can remember the security PIN you created five years ago.

The "I don't have time—I'll do it the long way" mindset is symptomatic of the lackluster conversational experiences users are accustomed to having with machines. But that's starting to change. One of the key elements of this convergence of technologies surrounding conversational AI involves intelligent and evolving ecosystems designed for accelerated automation powered by one of humankind's oldest adaptations: conversation. Make no mistake, it's not that conversational AI isn't going anywhere—it's going everywhere.

Groundbreaking as they've been, recent innovations like Alexa and Google Home hardly qualify as conversational AI. Asking smart speakers to issue weather reports, set a timer, or play a song are very limited and immature applications of conversational AI, though they hint at its nascent power. Smart speakers have completely upended the speaker industry, to the point where it might be difficult to find a new speaker for sale that doesn't have built-in conversational capabilities. But how powerful does a smart speaker become when it's not limited to the things that Siri or Alexa can do? What happens when you can ask your speaker to play "Mr. Roboto" by Styx and then follow up with another request: "I want to buy a copy of the book that Marc Maron mentioned during the intro to his podcast today. I don't remember the title but see if there's a copy available from Powell's before looking on Amazon."

What will happen is, a few minutes later, a text message could appear on your phone with a link showing a hardcover copy of Camera Man by Dana Stevens available on Powells.com. By replying, "Yes. Please buy" via text, you'd be communicating with the same interface that you initially spoke to—an umbrella conversational interface that has become your primary interaction point with most of the technology in your life. Once this scenario is possible, you won't think of technology in terms of different apps, because you'll rarely need to open and interact with an app. Domo arigato, Mr. Roboto.

Of course, human conversation is broader than the spoken word, as we have many ways of communicating our thoughts and needs. Humans frequently incorporate gestures, facial expressions, visual aids, and sounds in conversation. As such, conversational AI encompasses a full breadth of what I call "multi-turn" or "multimodal" interactions. Because they are part of an interconnected ecosystem, these multimodal interactions can leverage those massive stores of data we're continually creating—unearthing massive opportunities for personalization and precision.

To explain what I mean by "multi-turn" or "multimodal": having a text conversation with an invisible machine might include that machine showing you part of a video to illustrate a point. If you're asking it to analyze a spreadsheet or data, it can draw you a graph on-the-fly to help visualize data points. If the interaction is ongoing and you're

Jack Loechner, "90% of Today's Data Created in Two Years," MediaPost, quoting IBM Marketing Cloud, "10 Key Marketing Trends For 2017," December 22, 2016, https://www.mediapost.com/publications/article/291358/90-of-todays-data-created-in-two-years.html], but in many ways this wealth of information represents a failure. To say it's being poorly leveraged is a bit of an understatement, but all of this is about to change.

about to start driving, the interface can move to voice command. These multimodal experiences mirror normal parts of conversations between people, and that sophistication enables humans to wield technological functions and capabilities using our most natural interface. These micro UIs, as you might call them, are dialogue-driven and, just like human conversations, they can include all kinds of audio and visual aids, and even haptic cues.

You could never have an experience this seamless and efficient while digging through nested tabs or apps—and many of the world's leading companies are coming around to this fact. Salesforce didn't just acquire Slack. Their CEO has openly admitted that they are rebuilding their entire organization around Slack. Microsoft is doing similarly with Teams: they're betting that an integrated communication platform and a unifying conversational interface—one machine that connects to everything—will benefit customers, employees, and organizations in big ways.

When this level of natural conversation becomes the primary interface between machines and the humans using them, the machine becomes invisible as the interface disappears. This line of thinking should be familiar to most experience design practitioners. One of the hallmarks of successful experience design is an interface that gets out of the way. The further the interface recedes into the background during an experience, the more frictionless that experience becomes. This lightens a user's cognitive load and helps them to get what they need from the technology more effectively (though it also represents a massive amount of orchestration behind-the-scenes).

With conversational AI, interfacing with machines no longer requires that we adapt to the way they communicate, which dramatically reduces friction in our experiences with machines and software. Conversational AI will go anywhere and everywhere—meaning that invisible machines will be, for lack of a less grandiose term, omnipresent. You'll be able to turn to your phone, any nearby smart speaker, or any voice-enable appliance and enlist the help of an invisible machine. This ties into another element of this convergence, which involves sequencing technology so that it can react and adapt to individual situations. Invisible machines galore, connected to ecosystems built for optimized problem solving.

**Add it all up and you get something called hyperautomation.**

## HYPERAUTOMATION MINUS THE HYPE

Sequencing disruptive, advanced technologies to work in concert is something Gartner calls hyperautomation, and it's as intense as it sounds. Gartner coined the term in 2019; by their estimation, "technology is now on the cusp of moving beyond augmentation that replaces a human capability and into augmentation that creates superhuman capabilities."

Implementing a strategy for hyperautomation is a massive undertaking that requires cooperation from every department within your organization. But you're not helpless in the face of hyperautomation. You can't succeed by resigning yourself to hyperautomation. It's not something you have to succumb to. You can take charge, but first you have to surrender many of the old ways of doing things. Hyperautomating isn't a space for concrete plans, flexibility is the name of the game, and you need an open platform that gives you control of the tools and software you use when automating experiences.

OneReach.ai was founded more than ten years ago, with a dedication to breaking down conversational AI into patterns that can be sequenced to hyperautomate the work that needs to be done. I've spent the spent the majority of my life trying to improve the ways people communicate with machines and the ways machines communicate with humans. Hyperautomation represents a massive leap forward on both sides of this equation. I believe hyperautomation has the potential for disruption on a scale similar to the advent of the printing press, the industrial revolution, and the dawn of the computer age itself. What's different is that the rate of change will most likely be much faster.

Hyperautomation is really an application strategy that goes beyond the development of AI and into how you sequence it with other disruptive technologies to solve complex problems as part of an organization-wide experience strategy. As you'll discover in the Tools and Architecture section in this book, technology exists right now that can help you move toward hyperautomating business processes, workflows conversations, and tasks to offer better-than-human experiences (or what my team is calling BTHx)—but most companies haven't yet tapped into it.

As Harvard Business School professor Marco Iansiti said about hyperautomation: "A lot of people think of this as disruption, like the taxi industry . . . being disrupted by Uber. It's not disruption. . . . Firms have been designed with management and labor since the Industrial Revolution. This is a fundamental change in the means of production, and it's affecting every industry across the board."

It's common for organizations to focus on disruptive technology in myopic ways—in this case, seeing automation as a means to handle simple tasks on human terms, like automating a coffee maker so that it brews a fresh pot of coffee at 8:45 am. What if, instead, the coffee pot was part of a better-than-human experience that not only adjusts the time it brews coffee and the amount it brews, but also cross references company calendars and brews an extra-strong pot of coffee in anticipation of a client coming straight to the office from an international flight. Better yet, imagine a nimble financial institution streamlining operations and eliminating overhead by hyperautomating tasks like approving loans, conducting credit score checks, and dispensing financial advice. This is how China's Ant Financial Group operates. They are a hyperautomated powerhouse whose mobile payment service boasts more than 450 million active users. (Apple Pay sits at about 12 million.)

When we think about AI we often think about the notion of singularity—the hypothetical point in time when a powerful superintelligence will surge past all human intelligence. There's also the notion of machines gaining General Intelligence and, thus, the ability to learn any intellectual task that humans can. These versions of superintelligence won't be the product of some super algorithm.

The perception of singularity or General Intelligence is more likely to emerge from an ecosystem of algorithms and technologies sequenced in intelligent ways to work in concert, and likely would be made up of contributions from different pieces of software engineered all over the world. These ecosystems of algorithms and sequenced technologies are very similar to the ones I'll describe in this book.

While the arrival of singularity could be many decades away, we've already quietly passed a significant milestone. Users are now having experiences with some conversational AI that are far more rewarding than what their human counterparts offer (BTHx). For evidence, I'll turn again to Ant Financial, who told MIT Technology Review that customer satisfaction with their chatbots had surpassed human performance. "There are many, many chatbot companies in Silicon Valley," Yuan Qi, Ant's Vice President & AI Chief Scientist, noted at the time. "We are the only one that can say, confidently, [our chatbots perform] better than human beings."

There's also the example of Lemonade, a tech-minded startup that has disrupted the rental insurance market with low prices, a program for donating premium surplus money to charity, and a better-than-human customer-facing conversational AI.

"The biggest thing that pushed me to convert to Lemonade was the utterly charming AI chatbot," Juliette van Winden wrote in a Medium post dedicated to their chatbot, Maya. "24/7, 365, day or night, Maya is there to answer any questions to guide the user through the sign-up process. Unlike the drag of signing up with other providers, it took me a total of two minutes to walk through all the steps with Maya. . . . What intrigued me the most, is that it didn't feel like I was chatting with a bot. Maya is funny and charismatic—which made the exchange feel authentic."

We'll explore all sorts of scenarios in the Lessons in Hyperautomation section of this chapter. But for now, imagine this kind of reality playing out in another context: Your router goes down. You place a call to your service provider and are guided through all the necessary system checks quickly and elegantly by a conversational app. Or, better yet, how about their hyperautomated ecosystem detects that your router is down and their conversational app reaches out to you before you even notice that you're offline. This is the work of an Intelligent Digital Worker, or IDW. Far more capable than a chatbot, the IDW is connected to this service provider's ecosystem, which is a network of interdependent technologies, processes, and people. This customer-service-oriented IDW reaches out to you because it got an alert from a fellow machine in the maintenance department that your location had lost coverage. The IDW is programmed to isolate and troubleshoot the issue by running background tasks while it speaks with you to verify your account and location. This intelligent digital worker could ask you to send a photo of the blinking lights on your router while it simultaneously looks internally at your connection status. It assesses and course-corrects in seconds and your router is back up and running inside of five minutes. Best part? You didn't have to wait on hold for a human operator, because there are unlimited digital agents at the ready—or because they called you, having detected the issue before you did. After experiencing this kind of BTHx firsthand, you'll never want to go back.

It's always been frustrating for us to watch organizations invest heavily in marketing to get customers in the door and then invest more money in call centers to get them out the door. Hyperautomation can close that gap and make every interaction a customer has with an organization pleasant and fruitful. Literally every interaction with a customer becomes an opportunity.

As conversational AI is sequenced with other technologies to contextualize massive amounts of data within an ecosystem that can give customers and employees access to elevated problem-solving capabilities, the world as we know it will change fundamentally.



https://www.technologyreview.com/2017/06/16/151178/ant-financial-chinas-giant-of-mobile-payments-is-rethinking-finance-with-ai/]

**In this book we'll explore the components of a robust ecosystem for hyperautomating business processes, workflows, tasks, and communications, along with what a strategy looks like for evolving these ecosystems.**

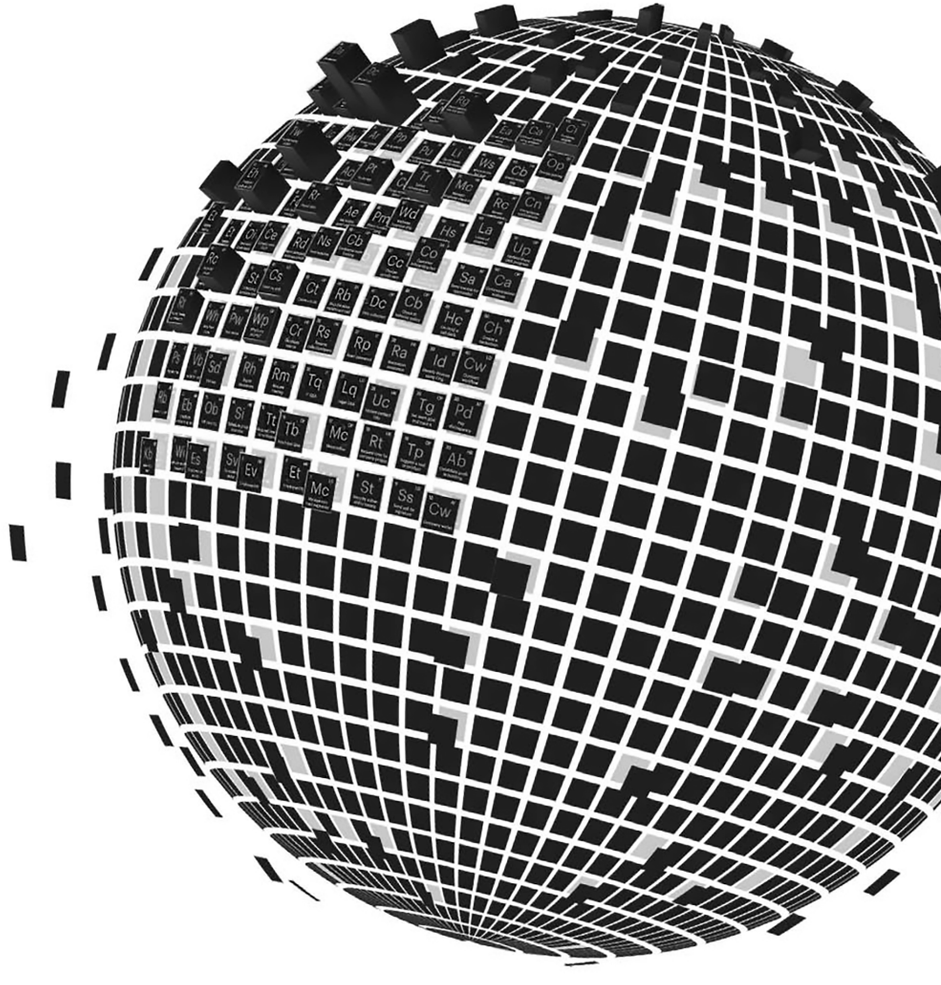
What's important to remember throughout is that the potential created by hyperautomation is so vast that the marketplace advantage can be staggering for companies with these ecosystems already in place. Making the most of this hyperdisruptive moment in history (and not being left behind) requires a holistic undertaking that touches on all aspects of your business. Random acts of technology—like deploying disparate machines that exist in isolation—will underwhelm your workforce and customers, leading to low adoption rates. A fully integrated approach, however, can bring about a totally new paradigm of productivity with unprecedented potential.

## "Wow, This Sounds Really Hard"

Hyperautomation is indeed a momentous undertaking. The easiest way to get started is often to automate internally first; start small by automating individual tasks and skills, not entire jobs. The simpler you make your starting point, the sooner you can test and iterate. The sooner you test and iterate, the sooner you can roll out an internal solution. You'll continue testing and iterating on that solution, using the momentum to find new skills to develop, test, iterate on, and deliver. You'll fumble often as you grow legs, but that's part of the process, too. In the realm of hyperautomation, we are more agile than agile. With the right tools and budding ecosystem, the iteration process becomes so speedy that failures are often quick rewards that point to better solutions. Because fixes and new solutions can be tested and deployed quickly and at will, your organization can build on wins and gain speed.

"Iteration" is a term tossed about freely in most enterprise settings, but it's important to remember that the real goal is to be continuously improving. By rolling out internal successes with automation and continually improving on them, you're both demonstrating to everyone in your organization, top to bottom, the process by which hyperautomation will take place—and introducing them to the ecosystem they will eventually call home. In the long-term, this will bring you closer to creating and testing customer-facing conversational applications. In the near-term, by focusing on helping your team members accomplish more and experience more satisfaction with their jobs, your customers will be rewarded downstream. This isn't the only way to reach the goal of hyperautomation, but it's generally the fastest way to accelerate your path to AI adoption on an organizational level.

**FIG 1.1. AN ECOSYSTEM OF INTELLIGENT DIGITAL WORKERS**



https://medium.com/marketing-in-the-age-of-digital/love-at-first-chat-with-lemonades-ai-chatbot-maya-7b4a105824bd]

# HYPERAUTOMATION BRINGS HYPERDISRUPTION

I can't help but reiterate that these changes to business are underway and they're going to accelerate in astonishing ways. The "hyper" is there for a reason, and the sequencing of disruptive technologies inside ecosystems built for hyperautomation will unleash hyperdisruptions across all industries, emerging suddenly and in ways that we won't always be able to predict. But while the hyper aspects of these disruptions are new, the sequencing of technology to solve complex problems has been with us for centuries.

The printing press is an easy disruption to point to, as it spread information all across medieval Europe and eventually the world in a truly revolutionary manner. But the printing press itself was a variation on presses that were already being used to squeeze grapes for wine and olives for oil. When Gutenberg printed his bible in 1455, it was the product of an orchestration of variations on other disruptive technologies: an oil-based improvement on existing inks, customizations to paper and moveable type originally developed in China, metallurgy, and the press.[NOTE: Jared Diamond, Guns, Germs, and Steel: The Fates of Human Societies (New York: W. W. Norton, 1997), 259.]

It's in our nature to innovate by orchestrating technologies in our favor, and hyperautomation represents a new era wherein anyone with an idea can have a voice. And with that voice, they can orchestrate disruptive technologies to accomplish things others haven't even dreamed of.

As I'll describe in these pages, assembling a diverse team and fostering a culture that champions change are keys to success in a landscape that will be continually disrupted. Embracing change lets you use speed and iteration to offset any fear of failure. Sometimes the best way to learn and make progress is to just start—adopting a practice of failing forward fast. Big changes on top of big changes are on the way, so alarm, urgency, and action are all merited. For those just getting started or who are already up to their necks, rest assured, there are practical ways to achieve hyperautomation.

Printing press–level disruptions will come by the week, not the decade. To help set your bearings, in chapter 5 we'll explore several scenarios for how hyperautomation might rewrite our experiences with technology and change our daily existence in a flash. But first, let's set the record straight on what conversational AI is, and isn't.

**KEY TAKEAWAYS**

- Hyperautomation is here, and it's already being used by companies like Ant and Lemonade to disrupt two long-standing industries: banking and insurance.
- Once the wider adoption of conversational AI takes hold, our interactions with machines will change fundamentally, ushering in an era of unprecedented technological advancement.
- Hyperautomation isn't something you have to succumb to. You can take charge, but you'll also have to sacrifice many of the old ways of doing things.
- In a world where organizations of all shapes and sizes are hyperautomating, orchestrating disruptive technologies to solve increasingly complex problems will be the order of the day.
- The accelerated orchestration of disruptive forces will create an atmosphere of hyperdisruption, where entire industries will be reimagined on a regular basis.

CHAPTER TWO

# WHAT CONVERSATIONAL ARTIFICIAL INTELLIGENCE IS—AND ISN'T

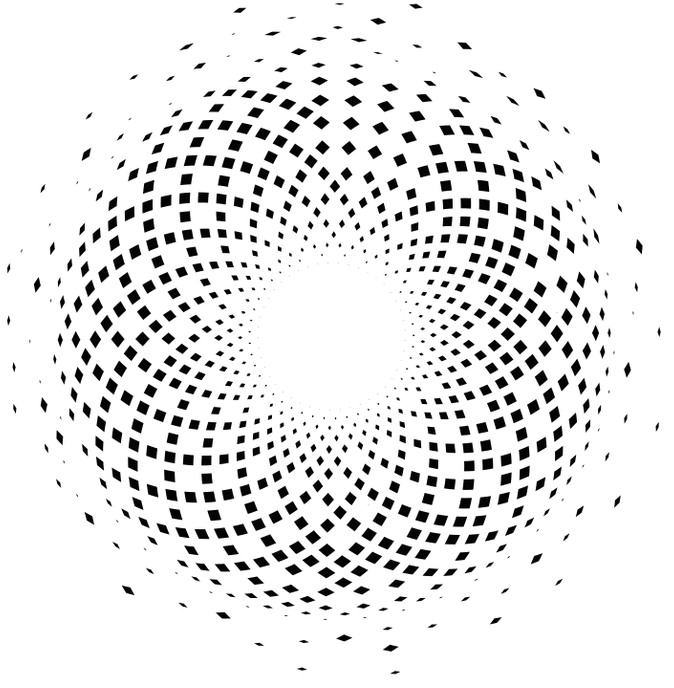
# WHAT CONVERSATIONAL ARTIFICIAL INTELLIGENCE IS—AND ISN'T

The resistance to automation and conversational AI has many facets. Much of the push-back relates to the automation of tasks that people don't believe machines can do well, or don't trust machines to do at all. As conversational AI and ecosystems for hyperautomating become the norm, however, it will become readily apparent that there are a vast number of tasks and processes that can be automated with great success by machines, so long as those machines have humans in control and guiding the process.

"Automation" refers to tasks humans typically perform being performed instead by machines. By extension, "hyperautomation" is achieved by successfully orchestrating of advanced technologies—like artificial intelligence, machine learning, natural language understanding, computer vision, conversational technology, and code-free development tools, and others—in order to automate tasks and processes that are outside the capability of humans alone. It's the coordination of advanced technologies to work in concert; to automate with massively enhanced impact.

It's tempting to think of conversational AI as nothing more than a new interface that experience designers can apply their skills to, but it's something much larger than that. Conversational AI represents a vast, emergent set of technologies obscured by layers of hype and misinformation. There's a very reasonable sense of urgency surrounding its adoption and implementation within the business community, but putting it to work requires a fuller understanding of what it actually entails.

In reality, successful implementation of conversational AI requires an open ecosystem built for hyperautomation, where a shared library of information and code-free design tools make high-level automation and continual evolution an everyday thing. While it's possible that this ecosystem can be built using technologies that are part of your existing ecosystem, a sizable evolution will be necessary nonetheless. With hyperautomation your ecosystem moves from being app-based, and limited in scalability by disparate graphical user interfaces—to skills- and functionality-based, with a unifying conversational interface connecting everything. In order to achieve hyperautomation, interchangeable technologies are sequenced



and orchestrated using an open platform that gives you both the freedom to implement the best functionalities from any vendor and the flexibility to iterate on solutions quickly and at-will. It's also important that you're able to orchestrate quickly and make continual improvements to those orchestrations. This is really about strategically sequencing technologies inside your ecosystem; hyperautomating means you're doing this quickly while constantly implementing iterative improvements.

When you start thinking about conversational AI in the right way, you begin to see that it's not a piece of technology to be wielded; it's part of a business strategy for building an ecosystem where various technologies are sequenced to automate tasks and processes in ways that surpass what humans alone are capable of. Conversational AI isn't magic, it's just math + logic—and when it's orchestrated properly, it eliminates all communication barriers between humans and machines.

Perhaps the easiest way to illustrate what conversational AI is would be to clarify what it is not. To follow are the truths behind some of the more persistent myths about conversational AI.

# DISPELLING COMMON MYTHS ABOUT CONVERSATIONAL ARTIFICIAL INTELLIGENCE

**Myth 1. Conversational AI is just talking to machines.**

Not even close. Natural Language Processing (NLP) and Natural Language Understanding (NLU) are actually just two of the smaller pieces of the conversational AI puzzle. Getting computers to the point where they can (a) contextualize the things that people say (NLP) as well as (b) understand the intent behind what's being said and (c) provide useful responses (NLU) required massive leaps in technology, touching on intent recognition, entity recognition, fulfillment, voice-optimized responses, dynamic text-to-speech, machine learning, and contextual awareness. At present, only a handful of organizations are leveraging this technology at high levels, and not just because experiences in a state of hyperautomation go beyond the limits of talking, typing—incorporating gestures, facial expressions, visual aids, sounds, and haptic feedback. Behind the scenes, conversational AI relies on a whole stew of technologies and processes, including integrations, task automation, multichannel optimization, conversational design, maintenance and optimization, and real-time analytics and reporting. In that context, comparing NLP/NLU to conversational AI is like comparing a bicycle wheel to an automobile. (The same can be said of text-to-speech (TTS) or "read aloud" technology that converts typed words into audio; and automated speech recognition (ASR), which allows users to speak commands rather than press numbers on a keypad.) In fact, the more complex a use case becomes, the less prominently NLP/NLU factors in. A Q&A bot addressing employee healthcare options is entirely NLP/NLU (e.g., identify question, provide correct answer, done). When automating the process of signing an employee up for the right health plan, however, NLP/NLU plays a limited role. Once it has helped identify that a user wants to sign up for a plan, other technologies and processes swing into action: authentication, gathering personal employee data, generating suggested plans, signing up users for the chosen plan, confirming transactions via email, etc. All of these actions require an ecosystem created for automation. Conversational AI is lightyears beyond just talking to machines.

**Myth 2. Conversational AI is an Add-On**.

Nope. Conversational AI isn't something you plug in and watch spring to life. Extracting value from conversational AI requires a comprehensive strategy, especially where legacy systems are involved. The good news is that the amount of work and introspection it takes to successfully implement and scale a conversational AI strategy pays dividends in the ways it unifies and improves all facets of an organization.

**Myth 3. The goal of conversational AI is to mimic humans.**

Humans are capable of amazing things—as a species, we're continually pushing the boundaries of what we can accomplish. But conversational AI isn't human; we're not trying to pretend it's human, and you shouldn't try to make it human. The point of conversational AI and hyperautomation is to unlock the ways sequenced technology can perform services far better than what humans alone are capable of.

**Myth 4. This stuff is easy.**

It can be challenging to integrate conversational AI into organizational operations. As much as we might be tempted to hurl chatbots at isolated workflows, in reality successful implementation of conversational AI requires an ecosystem built for hyperautomation—where a shared library of information, patterns, and templates join with code-free design tools to produce high-level automation and continual evolution.

**Myth 5. You should start big.**

Most organizations don't need a large-scale, public-facing deployment—like Bank of America's virtual financial assistant, Erica—to put conversational AI to good use. It's often more effective to begin by working within your organization, automating internal tasks with help from the people who understand those tasks best. Working internally, you can also get a handle on how to sequence technology to complete jobs in more efficient and rewarding ways than humans alone are able to. This will form the groundwork for an ecosystem of conversational AI that can grow to include customers at a point when you're equipped to provide them with optimized experiences. You'll want to start small, simple, and internally—then iteratively advance, expand, and scale.

**Myth 6. Business-minded people know enough to pick the best conversational AI platforms.**

Given the complexity I've already discussed (not to mention what's to come in subsequent chapters), the business minds in your organization can't make conversational AI decisions alone. This calls for a collaborative effort that draws on expertise from various disciplines within your organization. You'll need input from those who understand the technology you currently have in place, those who have relationships with your customer base, and those who understand the aspects of your business that are ripe for automation. The best way to make sure you acquire the right tools and systems is to develop a fuller understanding of what goes into creating an ecosystem for hyperautomation to live in. Once you have a shared vision of what you want to build, it will be far easier to see which products will truly provide you with valuable solutions. Successful conversational AI platforms emerge from a collaborative effort among key personnel in your organization.

**Myth 7. You can do this alone.**

Hyperautomation isn't an initiative you can put solely in the hands of any one group in your organization. You need a core team of specialists instituting a process that involves members from every department in your organization. That core team can come from within your organization, or the outside world, but its members have to be adept enablers. As they work their own skill-specific roles, they need to always be evangelizing the potential for anyone with a good idea to contribute to the design of automated solutions. Hyperautomation is an all-hands-on-deck journey.

**Myth 8. To get conversational AI right you need to hire an expert.**

The sequencing of conversational AI in ecosystems built for hyperautomating is still in the inception stage, so there aren't yet expert agencies or hired guns that can get you up-to-speed with the flip of a switch. That's not to say that there aren't great partners out there who can help you with elements of hyperautomation. Just note that having others set up everything for you isn't the best use of your investment. Instead, select a partner committed to training your own team in successful implementation methods.

**Myth 9. There is one platform to rule them all.**

Giant companies like Amazon, Microsoft, and IBM have very sophisticated products that address certain facets of conversational AI, but none of them has a platform or service that can deliver hyperautomation. What you need is an open system that allows you to leverage a whole host of tools, enabling you to sequence and orchestrate different technologies. Hyperautomation requires a network of elements working together in an evolutionary fashion. To quickly grow your ecosystem, you'll want to try as many solutions as possible as quickly as possible. The best approach is to iterate rapidly—to continually improve by trying out new configurations and isolating the best tools, AIs, and algorithms for your business. Open systems allow you to understand, analyze, and manage the relationships among the moving parts inside your ecosystem, which is crucial when moving toward hyperautomation. Remember, this isn't a race to adopt any specific technology; it's a race to put yourself in the position to adopt as many different technologies as possible.

**Myth 10. Phones aren't relevant to hyperautomating.**

It's easy to think of conversational AI as a web-based technology, but telephony is where most of its cost-saving attributes are found. While a sizeable number of people spend the majority of their time sitting at a computer, just about everyone carries a smaller computer on their person, all day long. Smartphones have all sorts of capabilities like SMS, voice, GPS, and banking that can be leveraged across ecosystems built for sequencing conversational AI.

There's also a large segment of the world's population whose only access to the internet comes through the less-expensive feature phone or the hybrid smart feature phone. According to Yiwen Wu a senior analyst at Strategy Analytics, "We estimate the global smartphone user base has risen dramatically from just 30k people in 1994 to 1.00 billion in 2012, and a record 3.95 billion today in June 2021. With an estimated 7.9 billion people in total on the planet in June 2021, it means 50% of the whole world now owns a smartphone."

Also, though call centers are declining in general, they are still very much a viable piece of many businesses around the world. Merijn te Booij—chief marketing officer of Genesys, a contact center technology company—told Vox how, in the early days of the pandemic, they "saw a lot of people moving away from digital channels very quickly to the voice channel, trying to get certainty, assurances, empathy, making sure that they got commitments on the enterprise decision."The bottom line is that phones of varying degrees of sophistication are key interface points for ecosystems where conversational AI is being sequenced. They can offer a broader reach and enhanced flexibility, so don't count them out.

[NOTE: "Strategy Analytics: Half the World Owns a Smartphone," Business Wire, June 24, 2021, 02:10 PM EST, https://www.businesswire.com/news/home/20210624005926/en/Strategy-Analytics-Half-the-World-Owns-a-Smartphone.]

[NOTE: Hilary George-Parkin, "One nation, on hold," Vox, May 13, 2020, 7:10am EDT, https://www.vox.com/the-goods/2020/5/13/21243420/call-centers-on-hold-customer-service-unemployment-airline-cable.]

**Myth 11. You can get to the next level with the system and pre-trained bots you already have.**

Most organizations that have made attempts at automation via conversational AI are stuck watching disparate chatbots sputter along independently without any meaningful connections to their fellow bots or the organization using them. This is a waste of time and resources that alienates customers and team members alike. For context, Gartner charts most of these out-of-box machines as being both low effort and low value. As you move toward higher effort/value integrations, your capabilities surge from being domain specific with limited channels and response capabilities, to machines that can function more like virtual assistants—capable of data-driven decision-making, working across channels, and behaving proactively. Getting to the much higher level of hyperautomation requires a strategy for building an ecosystem in which this kind of conversational AI thrives. Through the process you might discover that certain elements of your existing ecosystem can become integrated into your evolved ecosystem, but they will never serve as the springboard for the necessary evolution.

Acquiring chatbots that are already set up to automate particular workflows your organization uses might seem like a shortcut to hyperautomation, but it will cost you more time in the long-term. As you automate workflows, you're going to find opportunities to improve them. If your only option is to query the machine's developers with your requests, you're entering a cycle of waiting: waiting on iterations, testing them, and then waiting for updates. This wastes time and kills momentum. You need customizable tools that allow you to make code-free design changes at will—changes you can then test internally and implement quickly. That's the true path to hyperautomation.

**Myth 12. You'll improve operations by automating existing workflows.**

On the surface, it might seem like the biggest win of conversational AI would be to have it run monotonous tasks in place of humans. While there is some usefulness in that, the real value comes with automating better ways of doing things. For example, let's say you call the IRS following up on a letter they sent you. Your first hurdle is to figure out which of the unintelligible clusters of voice-automated options most closely applies to your situation. Then you repeat that process a few more times as you move through murky layers of their phone tree, unsure if you're headed to the right department and expecting to wait on hold for hours for find out. What if, instead, you were greeted by an Intelligent Digital Worker that could verify your personal information while cross-referencing your recent tax history to infer that you're calling about a letter that was sent last week. What if that IDW could also tell you that the payment you sent was received after the letter was sent, give you a confirmation number, and send you on your way in under five minutes? Within an ecosystem of integrated conversational AI that kind of better-than-human experience can readily be designed and implemented.

**Myth 13. This is like any other software buildout.**

The waterfall approach to software creation is severely outdated and has sprung countless leaks over the years. If you attempt to apply it within the framework of designing conversational AI solutions, you will drown. Creating an ecosystem for hyperautomation using conversational AI is an iterative process that demands the flexibility of rapid, code-free design tools for the steady deployment of new solutions. Even companies that are used to agile methodologies should prepare for faster iteration cycles than they might be accustomed to.

**Myth 14. You can upload all your organization's stored data and proceed to launch.**

If only it were that simple. Accumulating massive amounts of data about your customers and business has never been easier, but extracting the value buried in the heaps is a complex and ongoing endeavor that requires a comprehensive strategy. (And though some may be tempted to address this concern with plug-and-play approach, note that these methods peak at about 40% accuracy, which means most of your data would be wasted.) In order to extract all its value, data needs to be categorized within a framework that makes it an active resource accessible across your entire ecosystem. One of the most potent tools for accelerating these efforts is a scenario where machines query humans within an organization whenever they bump up against the limits of their abilities. Not only can the humans help move the current interaction forward, they can also instruct the machines so that the machines can solve the problem without help in the future.

**KEY TAKEAWAYS:**

- As conversational AI becomes commonplace, a vast number of tasks and processes will be automated with great success by machines guided and controlled by humans.
- Success with conversational AI requires an open ecosystem built for hyperautomation, where high-level automation and continual evolution occur daily.
- When orchestrated properly, conversational AI eliminates all communication barriers between humans and machines.
- Before getting started with conversational AI, it's critical to let go of the many common myths associated with it.

# ABOUT THE AUTHORS

**Robb Wilson**

Robb Wilson is the founder, lead designer, and chief technologist behind OneReach.ai, the highest-scoring company in Gartner's first Critical Capabilities for Enterprise Conversational AI Platforms report and a leader for Completeness of Vision and Ability to Execute in Gartner's inaugural 2022 Magic Quadrant for Enterprise Conversational AI Platforms. Raised under the tutelage of philosopher Marshall McLuhan—who predicted the internet 30 years before it became reality—Robb has spent more than two decades applying his deep understanding of user-centric design to unlocking hyperautomation. He built UX Magazine into the world's largest experience design publication while simultaneously creating Effective UI, a full-service UX firm that competed with IDEO and Frog Design. In addition to launching 15 startups and collecting over 130 awards across the fields of design and technology, Robb has held executive roles at several publicly traded companies and mentored colleagues who went on to leadership roles at Amazon Alexa, Google, Ogilvy, GE, Salesforce, Instagram, LinkedIn, Disney, Microsoft, Mastercard, and Boeing. Robb puts the same passion into building a surfboard and renovating his home that he instills in the startups he routinely bootstraps without venture or third-party capital. A trusted thought leader in the realm of conversational AI and hyperautomation, Robb has played a part in creating a wide variety of products, apps, and movies that have touched nearly every person on the planet. Robb is also a humble husband, father, and grandfather, and reading this bio always makes him blush a bit.

**Josh Tyson**

Josh Tyson is an author and producer who has held leadership roles for a variety of organizations, including TEDxMileHigh and UX Magazine He is the co-cost of N9K, a comedy podcast from the future. His writing has appeared in numerous publications over the years, including Big Brother Skateboarding, Chicago Reader, FLAUNT, The New York Times, SLAP, Stop Smiling, Thrasher, and Westword. Josh lives with his family in Denver, where he enjoys skateboarding, yoga, and getting his kids to watch the weirdest films they can find on the Criterion Channel.